

1.2.1 Access Networks

Having considered the applications and end systems at the “edge of the network,” let’s next consider the access network—the network that physically connects an end system to the first router (also known as the “edge router”) on a path from the end system to any other distant end system. Figure 1.4 shows several types of access

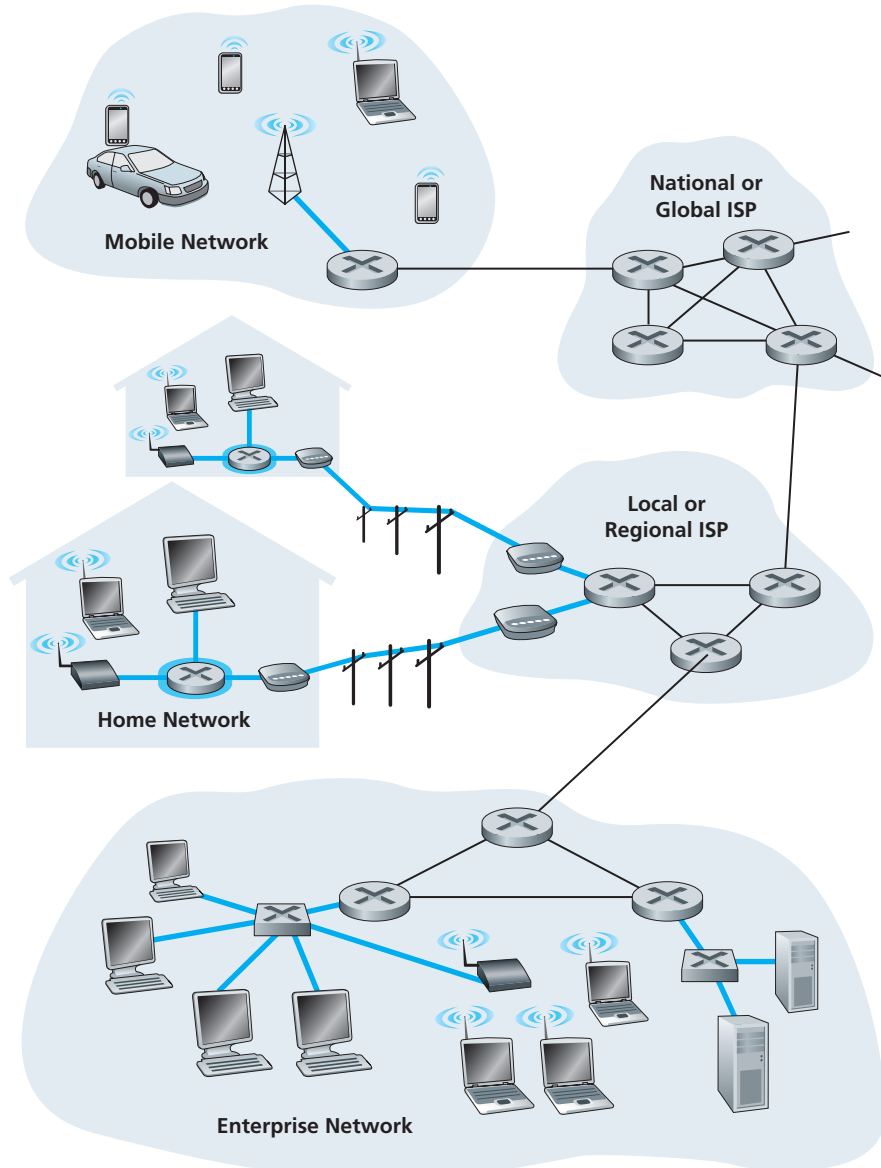


Figure 1.4 ♦ Access networks

networks with thick, shaded lines, and the settings (home, enterprise, and wide-area mobile wireless) in which they are used.

Home Access: DSL, Cable, FTTH, Dial-Up, and Satellite

In developed countries today, more than 65 percent of the households have Internet access, with Korea, Netherlands, Finland, and Sweden leading the way with more than 80 percent of households having Internet access, almost all via a high-speed broadband connection [ITU 2011]. Finland and Spain have recently declared high-speed Internet access to be a “legal right.” Given this intense interest in home access, let’s begin our overview of access networks by considering how homes connect to the Internet.

Today, the two most prevalent types of broadband residential access are **digital subscriber line (DSL)** and cable. A residence typically obtains DSL Internet access from the same local telephone company (telco) that provides its wired local phone access. Thus, when DSL is used, a customer’s telco is also its ISP. As shown in Figure 1.5, each customer’s DSL modem uses the existing telephone line (twisted-pair copper wire, which we’ll discuss in Section 1.2.2) to exchange data with a digital subscriber line access multiplexer (DSLAM) located in the telco’s local central office (CO). The home’s DSL modem takes digital data and translates it to high-frequency tones for transmission over telephone wires to the CO; the analog signals from many such houses are translated back into digital format at the DSLAM.

The residential telephone line carries both data and traditional telephone signals simultaneously, which are encoded at different frequencies:

- A high-speed downstream channel, in the 50 kHz to 1 MHz band
- A medium-speed upstream channel, in the 4 kHz to 50 kHz band
- An ordinary two-way telephone channel, in the 0 to 4 kHz band

This approach makes the single DSL link appear as if there were three separate links, so that a telephone call and an Internet connection can share the DSL link at the same time. (We’ll describe this technique of frequency-division multiplexing in

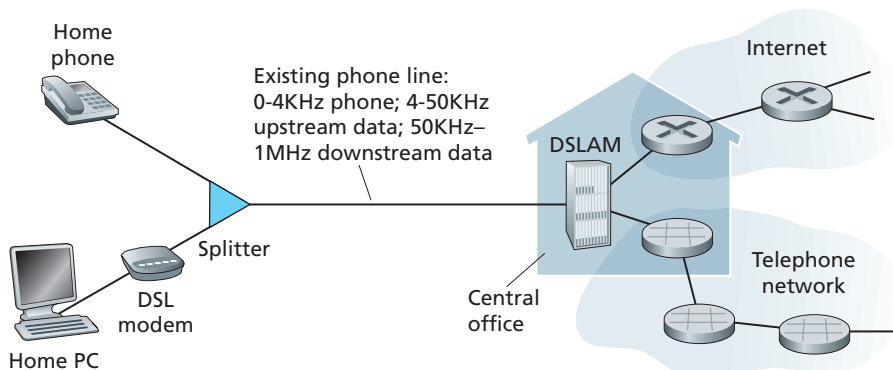


Figure 1.5 ♦ DSL Internet access

Section 1.3.1). On the customer side, a splitter separates the data and telephone signals arriving to the home and forwards the data signal to the DSL modem. On the telco side, in the CO, the DSLAM separates the data and phone signals and sends the data into the Internet. Hundreds or even thousands of households connect to a single DSLAM [Dischinger 2007].

The DSL standards define transmission rates of 12 Mbps downstream and 1.8 Mbps upstream [ITU 1999], and 24 Mbps downstream and 2.5 Mbps upstream [ITU 2003]. Because the downstream and upstream rates are different, the access is said to be asymmetric. The actual downstream and upstream transmission rates achieved may be less than the rates noted above, as the DSL provider may purposefully limit a residential rate when tiered service (different rates, available at different prices) are offered, or because the maximum rate can be limited by the distance between the home and the CO, the gauge of the twisted-pair line and the degree of electrical interference. Engineers have expressly designed DSL for short distances between the home and the CO; generally, if the residence is not located within 5 to 10 miles of the CO, the residence must resort to an alternative form of Internet access.

While DSL makes use of the telco's existing local telephone infrastructure, **cable Internet access** makes use of the cable television company's existing cable television infrastructure. A residence obtains cable Internet access from the same company that provides its cable television. As illustrated in Figure 1.6, fiber optics connect the cable head end to neighborhood-level junctions, from which traditional coaxial cable is then used to reach individual houses and apartments. Each neighborhood junction typically supports 500 to 5,000 homes. Because both fiber and coaxial cable are employed in this system, it is often referred to as hybrid fiber coax (HFC).

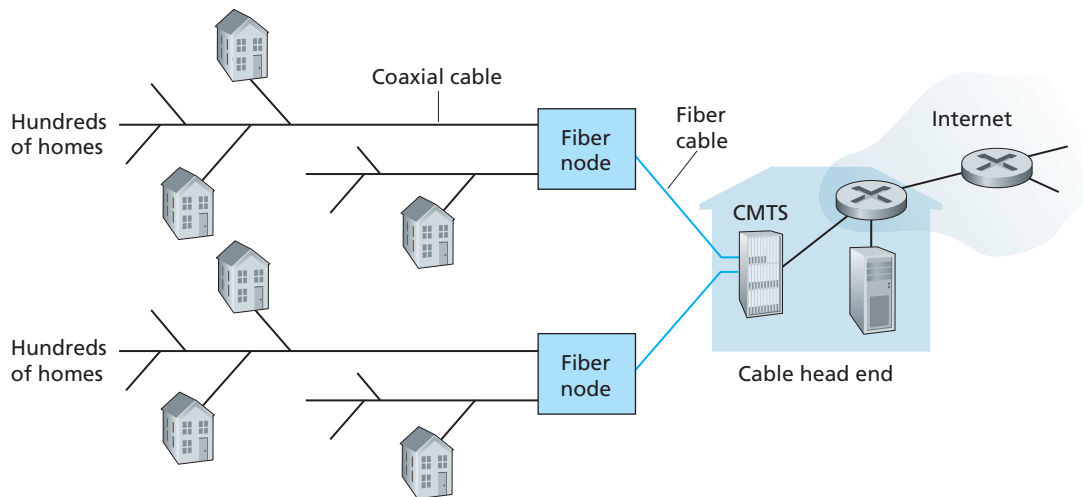


Figure 1.6 ♦ A hybrid fiber-coaxial access network

Cable internet access requires special modems, called cable modems. As with a DSL modem, the cable modem is typically an external device and connects to the home PC through an Ethernet port. (We will discuss Ethernet in great detail in Chapter 5.) At the cable head end, the cable modem termination system (CMTS) serves a similar function as the DSL network's DSLAM—turning the analog signal sent from the cable modems in many downstream homes back into digital format. Cable modems divide the HFC network into two channels, a downstream and an upstream channel. As with DSL, access is typically asymmetric, with the downstream channel typically allocated a higher transmission rate than the upstream channel. The DOCSIS 2.0 standard defines downstream rates up to 42.8 Mbps and upstream rates of up to 30.7 Mbps. As in the case of DSL networks, the maximum achievable rate may not be realized due to lower contracted data rates or media impairments.

One important characteristic of cable Internet access is that it is a shared broadcast medium. In particular, every packet sent by the head end travels downstream on every link to every home and every packet sent by a home travels on the upstream channel to the head end. For this reason, if several users are simultaneously downloading a video file on the downstream channel, the actual rate at which each user receives its video file will be significantly lower than the aggregate cable downstream rate. On the other hand, if there are only a few active users and they are all Web surfing, then each of the users may actually receive Web pages at the full cable downstream rate, because the users will rarely request a Web page at exactly the same time. Because the upstream channel is also shared, a distributed multiple access protocol is needed to coordinate transmissions and avoid collisions. (We'll discuss this collision issue in some detail in Chapter 5.)

Although DSL and cable networks currently represent more than 90 percent of residential broadband access in the United States, an up-and-coming technology that promises even higher speeds is the deployment of **fiber to the home (FTTH)** [FTTH Council 2011a]. As the name suggests, the FTTH concept is simple—provide an optical fiber path from the CO directly to the home. In the United States, Verizon has been particularly aggressive with FTTH with its FIOS service [Verizon FIOS 2012].

There are several competing technologies for optical distribution from the CO to the homes. The simplest optical distribution network is called direct fiber, with one fiber leaving the CO for each home. More commonly, each fiber leaving the central office is actually shared by many homes; it is not until the fiber gets relatively close to the homes that it is split into individual customer-specific fibers. There are two competing optical-distribution network architectures that perform this splitting: active optical networks (AONs) and passive optical networks (PONs). AON is essentially switched Ethernet, which is discussed in Chapter 5.

Here, we briefly discuss PON, which is used in Verizon's FIOS service. Figure 1.7 shows FTTH using the PON distribution architecture. Each home has

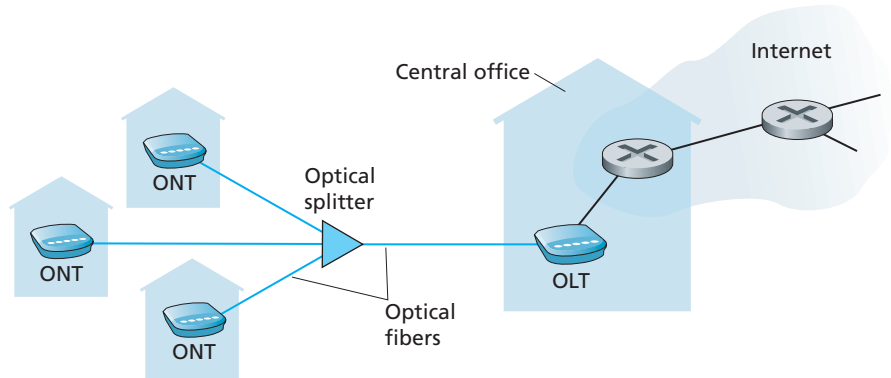


Figure 1.7 ♦ FTTH Internet access

an optical network terminator (ONT), which is connected by dedicated optical fiber to a neighborhood splitter. The splitter combines a number of homes (typically less than 100) onto a single, shared optical fiber, which connects to an optical line terminator (OLT) in the telco's CO. The OLT, providing conversion between optical and electrical signals, connects to the Internet via a telco router. In the home, users connect a home router (typically a wireless router) to the ONT and access the Internet via this home router. In the PON architecture, all packets sent from OLT to the splitter are replicated at the splitter (similar to a cable head end).

FTTH can potentially provide Internet access rates in the gigabits per second range. However, most FTTH ISPs provide different rate offerings, with the higher rates naturally costing more money. The average downstream speed of US FTTH customers was approximately 20 Mbps in 2011 (compared with 13 Mbps for cable access networks and less than 5 Mbps for DSL) [FTTH Council 2011b].

Two other access network technologies are also used to provide Internet access to the home. In locations where DSL, cable, and FTTH are not available (e.g., in some rural settings), a satellite link can be used to connect a residence to the Internet at speeds of more than 1 Mbps; StarBand and HughesNet are two such satellite access providers. Dial-up access over traditional phone lines is based on the same model as DSL—a home modem connects over a phone line to a modem in the ISP. Compared with DSL and other broadband access networks, dial-up access is excruciatingly slow at 56 kbps.

Access in the Enterprise (and the Home): Ethernet and WiFi

On corporate and university campuses, and increasingly in home settings, a local area network (LAN) is used to connect an end system to the edge router. Although there are many types of LAN technologies, Ethernet is by far the most prevalent access technology in corporate, university, and home networks. As shown in Figure 1.8, Ethernet users use twisted-pair copper wire to connect to an Ethernet switch, a

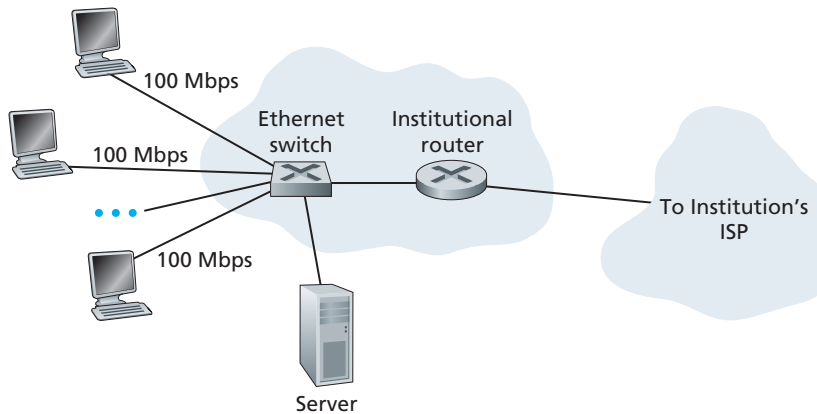


Figure 1.8 ♦ Ethernet Internet access

technology discussed in detail in Chapter 5. The Ethernet switch, or a network of such interconnected switches, is then in turn connected into the larger Internet. With Ethernet access, users typically have 100 Mbps access to the Ethernet switch, whereas servers may have 1 Gbps or even 10 Gbps access.

Increasingly, however, people are accessing the Internet wirelessly from laptops, smartphones, tablets, and other devices (see earlier sidebar on “A Dizzying Array of Devices”). In a wireless LAN setting, wireless users transmit/receive packets to/from an access point that is connected into the enterprise’s network (most likely including wired Ethernet), which in turn is connected to the wired Internet. A wireless LAN user must typically be within a few tens of meters of the access point. Wireless LAN access based on IEEE 802.11 technology, more colloquially known as WiFi, is now just about everywhere—universities, business offices, cafes, airports, homes, and even in airplanes. In many cities, one can stand on a street corner and be within range of ten or twenty base stations (for a browseable global map of 802.11 base stations that have been discovered and logged on a Web site by people who take great enjoyment in doing such things, see [wagle.net 2012]). As discussed in detail in Chapter 6, 802.11 today provides a shared transmission rate of up to 54 Mbps.

Even though Ethernet and WiFi access networks were initially deployed in enterprise (corporate, university) settings, they have recently become relatively common components of home networks. Many homes combine broadband residential access (that is, cable modems or DSL) with these inexpensive wireless LAN technologies to create powerful home networks [Edwards 2011]. Figure 1.9 shows a typical home network. This home network consists of a roaming laptop as well as a wired PC; a base station (the wireless access point), which communicates with the wireless PC; a cable modem, providing broadband access to the Internet; and a router, which interconnects the base station and the stationary PC with the cable modem. This network allows household members to have broadband access to the Internet with one member roaming from the kitchen to the backyard to the bedrooms.

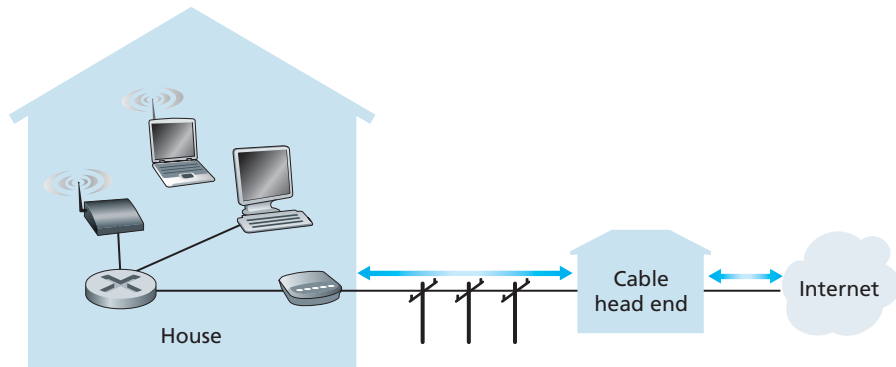


Figure 1.9 ♦ A typical home network

Wide-Area Wireless Access: 3G and LTE

Increasingly, devices such as iPhones, BlackBerrys, and Android devices are being used to send email, surf the Web, Tweet, and download music while on the run. These devices employ the same wireless infrastructure used for cellular telephony to send/receive packets through a base station that is operated by the cellular network provider. Unlike WiFi, a user need only be within a few tens of kilometers (as opposed to a few tens of meters) of the base station.

Telecommunications companies have made enormous investments in so-called third-generation (3G) wireless, which provides packet-switched wide-area wireless Internet access at speeds in excess of 1 Mbps. But even higher-speed wide-area access technologies—a fourth-generation (4G) of wide-area wireless networks—are already being deployed. LTE (for “Long-Term Evolution”—a candidate for Bad Acronym of the Year Award) has its roots in 3G technology, and can potentially achieve rates in excess of 10 Mbps. LTE downstream rates of many tens of Mbps have been reported in commercial deployments. We’ll cover the basic principles of wireless networks and mobility, as well as WiFi, 3G, and LTE technologies (and more!) in Chapter 6.

1.2.2 Physical Media

In the previous subsection, we gave an overview of some of the most important network access technologies in the Internet. As we described these technologies, we also indicated the physical media used. For example, we said that HFC uses a combination of fiber cable and coaxial cable. We said that DSL and Ethernet use copper wire. And we said that mobile access networks use the radio spectrum.